PRIME MINISTER'S OFFICE.
REGIONAL ADMINISTRATION AND LOCAL GOVERNME
ARUSHA DISTRICT COUNCIL.
FORM THREE TERMINAL EXAMINATIONS
BASIC MATHEMATICS.

MARKING SCHEME.

1(a)(i) Buying exercise books $= \frac{1}{3} \times 36,000$

$= 12,000$ TSHS. 01 Mark

(ii) Buying Calculator = ½ of remaining money

$= \frac{1}{2}(36,000 - 12000)$

$= \frac{1}{2} \times 24000$

$= 12000$ TSHS. 01 Mark

iii) Different expenses $= 36000 - 12000 - 12000$

$= 12000$ SHS. 01 Mark.

(b)(i) ~~Perce~~ Percentage of football $= \frac{52}{80} \times 100 =$

$= 65\%$. 1.5 marks

(ii) Percentage of netball $= 100\% - 65\%$

$= 35\%$. 1.5 marks.

2(a) $\frac{5+\sqrt{7}}{\sqrt{3}+\sqrt{7}} = \frac{5+\sqrt{7}}{\sqrt{3}+\sqrt{7}} \times \frac{\sqrt{3}-\sqrt{7}}{\sqrt{3}-\sqrt{7}}$

$= \frac{5\sqrt{3} - 5\sqrt{7} + \sqrt{21} - 7}{3 - 3\sqrt{7} + 3\sqrt{7} - 7}$

$= \frac{5\sqrt{3} - 5\sqrt{7} + \sqrt{21} - 7}{-4}$

$= \frac{5}{4}\sqrt{2} - \frac{5}{4}\sqrt{3} - \frac{\sqrt{21}}{4} + \frac{7}{4}$ 1.5 Marks.

2(b) $3\log_3 x - \log_3 24 + 4 = 0$

$4 + 3\log_3 x = \log_3 24 + 0$

$3\log_3 x = \log_3 24 - 4$

$\log_3 x^3 = \log_3 24 - \log_3 81$

$\log_3 x^3 = \log_3 \left(\frac{24}{81}\right)$

$x^3 = \frac{8}{27}$

$x^3 = \left(\frac{2}{3}\right)^3$

$x = \frac{2}{3}$

1·5 Marks

3 (a) Let $n(H)$ = number of student study History

$n(G)$ = number of student study Geography

$n(H \cap G)$ = no. of student study both.

$n(H \cup G) = n(H) + n(G) - n(H \cap G)$

$= 45 + 40 - 5$

$n(H \cup G) = 85 - 5 = 80$.

Students studying neither = No. of student in class $- n(H \cup G)$

$= 100 - 80$

$=$ 20 students · 01 Marks

b (i) $\mathcal{U} = \{15, 30, 45, 60, 75\}$.

$A = \{15, 45\}$

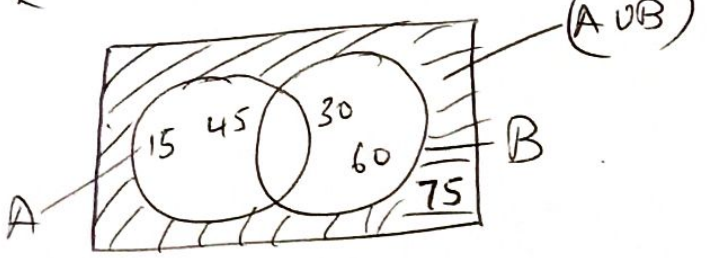
$B = \{30, 60\}$

$A \cup B = \{15, 45, 30, 60\}$

$\therefore (A \cup B)' = \{75\}$

01 Marks

(ii)



4 a) Distance from home to school = $2\frac{1}{2}$ km

Distance for three days = $2\frac{1}{2} \times 2$ (go and return) $\times 3$

$= \frac{5}{2} \times 2 \times 3$

$= 15$ km

$= 15000$ m 02 Marks.

b(i) For similar figures ratio of corresponding sides are equal and corresponding angles are equal.

$\therefore \angle AEB = \angle DEC$ (Vertical opp. angles)

$\angle EAB = \angle EDC$ (Right angle/given)

$\angle ABE = \angle DCE$ ($BA // DC$)

$\therefore \triangle AEB \sim \triangle CDE$ (Corresponding $\angle$, AAA)

02 Marks

(ii) $\frac{DC}{BA} = \frac{DE}{AE}$

$\frac{CD}{36} = \frac{32}{24}$

$CD = \frac{^{12}36 \times 32^{4}}{_{3}24} = 48$ m 02 Marks.

5 (a) Gradient (m) = $\frac{\text{Change in } y}{\text{Change in } x}$

$-2 = \frac{5+3}{k-2}$

$-2(k-2) = 8$

$-2k + 4 = 8$

$\frac{-2k}{-2} = \frac{4}{-2}$

$+k = -2$

$k = -2$ 03 Marks.

b) 1min = 500ml ⟶ 1min = 0.5l.

From 02:12pm to 05:04pm = 2 Hours and 52min

$= (2 \times 60) + 52$

$= 120 + 52$

$= 172$ Minutes.

Total amount of water for the time (in litres)

$= 172 \text{ minutes} \times 0.5\,l$.

$= \underline{86 \text{ litres}}$.

03 Marks

6 (a) $R\hat{P}Q = R\hat{Q}P$ (given)

$S\hat{P}Q = S\hat{Q}P$ (given)

$R\hat{Q}S = ?$

$S\hat{P}Q = S\hat{Q}P$ (Isosceles ΔPSQ) $= 34^\circ$

$RQP = 72^\circ$ given / Isosceles ΔPRQ.

$\therefore R\hat{Q}S = 72^\circ + 34$

$= 96^\circ$

03 Marks.

(b) Area of a rectangle $= L \times w = 72 \times 40$

$= 2880\ cm^2$

Area of a $\Delta = \frac{1}{2}bh$.

$2880 = \frac{1}{2} \times 60 \times h$

$\frac{2880}{3} = \frac{30h}{3}$

$96cm = h$

03 Marks.

$\therefore$ Height of the triangle $= \underline{96cm}$

7 (a) $80\% = 16000$

$100\% = ?$

$x = \frac{100 \times 16000}{80} = 20000/=$ 02 Marks

The original price is 20,000 shillings

(b) $V \propto kh$

$V = kh$

$\frac{140}{5} = \frac{k5}{5}$

$28 = k$

$V = kh$

$= 28 \times 1.5$

$= 42 cm^3$. 02 Marks

(c) Let $x = 0.1\dot{3}$ --- (1)

$10x = 13.\dot{3}$ --- (2)

$10x - x = 13.\dot{3} - 0.1\dot{3}$ --- (3)

$\frac{9x}{9} = \frac{1.2}{9}$

$x = \frac{1.2}{9}$

$x = \frac{12}{90}$

02 Marks.

8(a) Let x be the first number
$x+2$ be the second number.

$x+x+2=46$

$2x=46-2$

$\frac{2x}{2}=\frac{44}{2}$

$x=22$

The first number, $x=22$
The second number, $x+2=24$.

02 Marks

(b) Let x be price of a book
y be price of a pen.

$$\begin{cases} 4x+11y=1240 \\ 4x+3y=680 \end{cases}$$

$\frac{8y}{8}=\frac{560}{8}$

$y=70$

From $4x+11y=1240$

$4x+11(70)=1240$

$4x=1240-770$

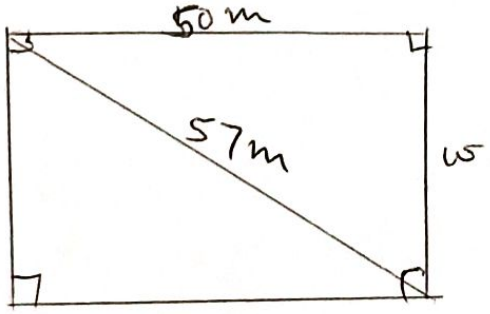
$\frac{4x}{4}=\frac{470}{4}$

$x=117.5$ shs

A book costs 117 and 50 cents
A pen costs 70 shs.

04 Marks.

9(a)



Let w be the width of the plot.

9(a) By applying Pyth. theorem

$w^2 + 50^2 = 57^2$

$w^2 = 57^2 - 50^2$

$w^2 = 3249 - 2500$

$w^2 = 749$

$w = \sqrt{749}$

$w = 27.37$

03 Marks

(b) $\frac{2\cos 60}{3} + \sin 30 + \frac{\sin 45}{\cos 45}$

$\frac{2}{3} \times 0.5 + 0.5 + 1$

$\frac{1}{3} + \frac{1}{2} + \frac{1}{1}$

$\frac{2+3+1}{6} = \frac{6}{6}$

$= 1$

03 Marks

10(a) $2 - x - x^2 = 0$

$-x^2 - x + 2 = 0$

$a = -1$

$b = -1$

$c = 2$

$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

$= \frac{1 \pm \sqrt{1 + 4 \times 1 \times 2}}{-2}$

$= \frac{1 \pm \sqrt{9}}{-2}$

$= \frac{1 \pm 3}{-2}$

3 Marks

$x = -2$ or $x = 1$

10(b) $T = h\sqrt{\frac{a-b}{d}}$

$T = h\left(\frac{a-b}{d}\right)^{\frac{1}{2}}$

$T^2 = h\left(\frac{a-b}{d}\right)^{\frac{1}{2}\times 2}$

$d \times T^2 = h\left(\frac{a-b}{d}\right) \times d$

$\frac{dT^2}{d} = \frac{h(a-b)}{T^2}$

$d = \frac{h(a-b)}{T^2}$

03 Marks

11 (a)

C.I.	f	c.f.	X	fx
40-49	2	2	44.5	89.0
50-59	4	6	54.5	218.0
60-69	7	13	64.5	451.5
70-79	9	22	74.5	670.5
80-89	5	27	84.5	422.5
90-99	3	30	94.5	283.5
				$\Sigma fx = 2135$.

Median class $= \frac{30}{2} = 15$ which is in 70-79.

Median $= L + \left(\frac{\frac{N}{2} - C_b}{W_a}\right) i$ where. 03 Marks.

L is the lower boundary of the median class

N is the total frequency.

C_b is the cumulative frequency below the median class

i is the class interval.

11(a) $\text{Median} = 70 + \frac{(15-13)}{9}10$

$= 70 + 2.222$

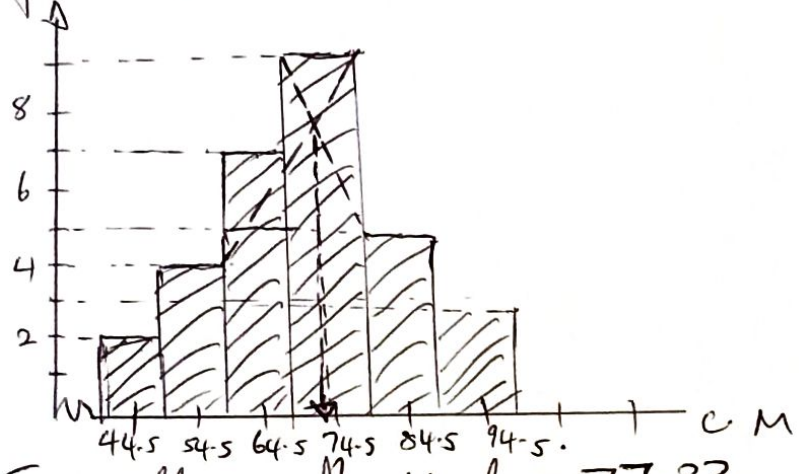
$\simeq 70 + 2.22$

$\simeq 72.22.$ 02 Marks

(b) $\text{Mean} = \frac{\sum fx}{N} = \frac{2135}{30} = 71.1667$

$\simeq \underline{71.17}$ 02 Marks

(c) Histogram



03 Marks

From the graph Mode $\simeq 77.33$

i.e.

$\text{Mode} = L + \left(\frac{t_1}{t_1+t_2}\right)i$ where

L = lower boundary of the modal class

t_1 is the excess of the modal frequency over the next lower class

t_2 is the excess of the modal frequency over the next higher class

i is the class interval.

$\text{Mode} = 70 + \left(\frac{2}{2+4}\right)10$

$= 70 + 3.33$

$\simeq 73.33.$

12(a) $A_n = A_1 + (n-1)d$

$A_{20} = A_1 + (20-1)d$

$A_{20} = A_1 + 19d$

$60 = A_1 + 19d$ ---- (I)

$A_{16} = A_1 + 15d$

$20 = A_1 + 15d$ ---- (II)

$\begin{cases} A_1 + 19d = 60 & ---(I) \\ A_1 + 15d = 20 & ---(II) \end{cases}$

$\frac{4d}{4} = \frac{40}{4}$

$d = 10$ ○ 4 Marks

Common difference is 10

b) $A_n = A_1 + (n-1)d$

$A_{20} = A_1 + 19d$

$60 = A_1 + 19(10)$ substituting $d = 10$.

$60 = A_1 + 190$

$A_1 = 60 - 190$

$A_1 = -130$ ○ 3 Marks

The first term is -130

c) $S_n = \frac{n}{2}[2A_1 + (n-1)d]$

$S_{30} = \frac{30}{2}[2(-130) + (30-1)10]$

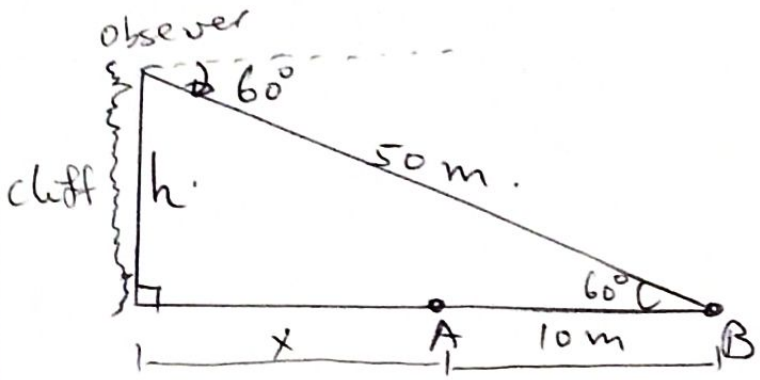
$= 15[-260 + 290]$

$= 15(30)$

$= 450$ ○ 3 Marks

The sum of the first 30 terms = 450

13(a)



$\sin 60^\circ = \frac{h}{10}$

$h = \sin 60 \times 50$

$h = 0.8660 \times 50$

$h = 43.3\,m$

The height of the cliff is 43.3m — 06 Marks

(b) $\cos 60^\circ = \frac{x+10}{50}$

$0.5 = \frac{x+10}{50}$

$x + 10 = 25$

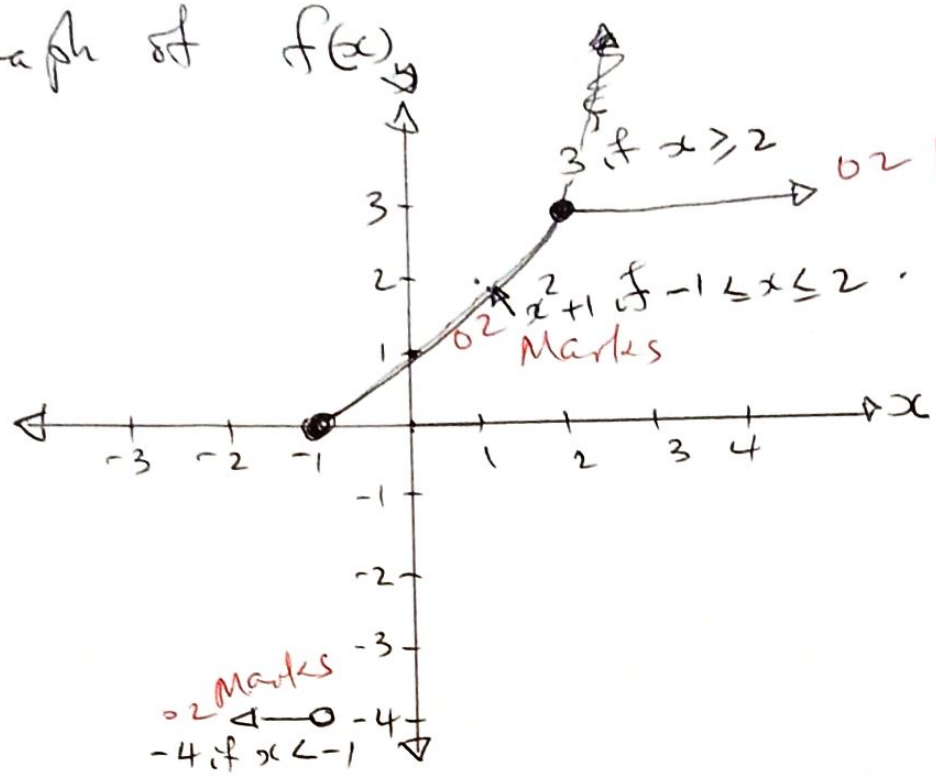
$x = 25 - 10$

$x = 15$

Distance of Boat, A to the base of the cliff is 15m — 03 Marks.

c) Distance of Boat B, to the base of the cliff is $(5+10)\,m = 25\,m$ — 01 Mark

14. (a) Graph of $f(x)$



b) Domain = {all real nos} 01 Mark

Range = $\{-4, 0, +1, +2, 3\}$ 01 Mark

c) Not one-to-one. Reason Some x values map to same y values. 01 Mark 01 Mark.

— V — X —